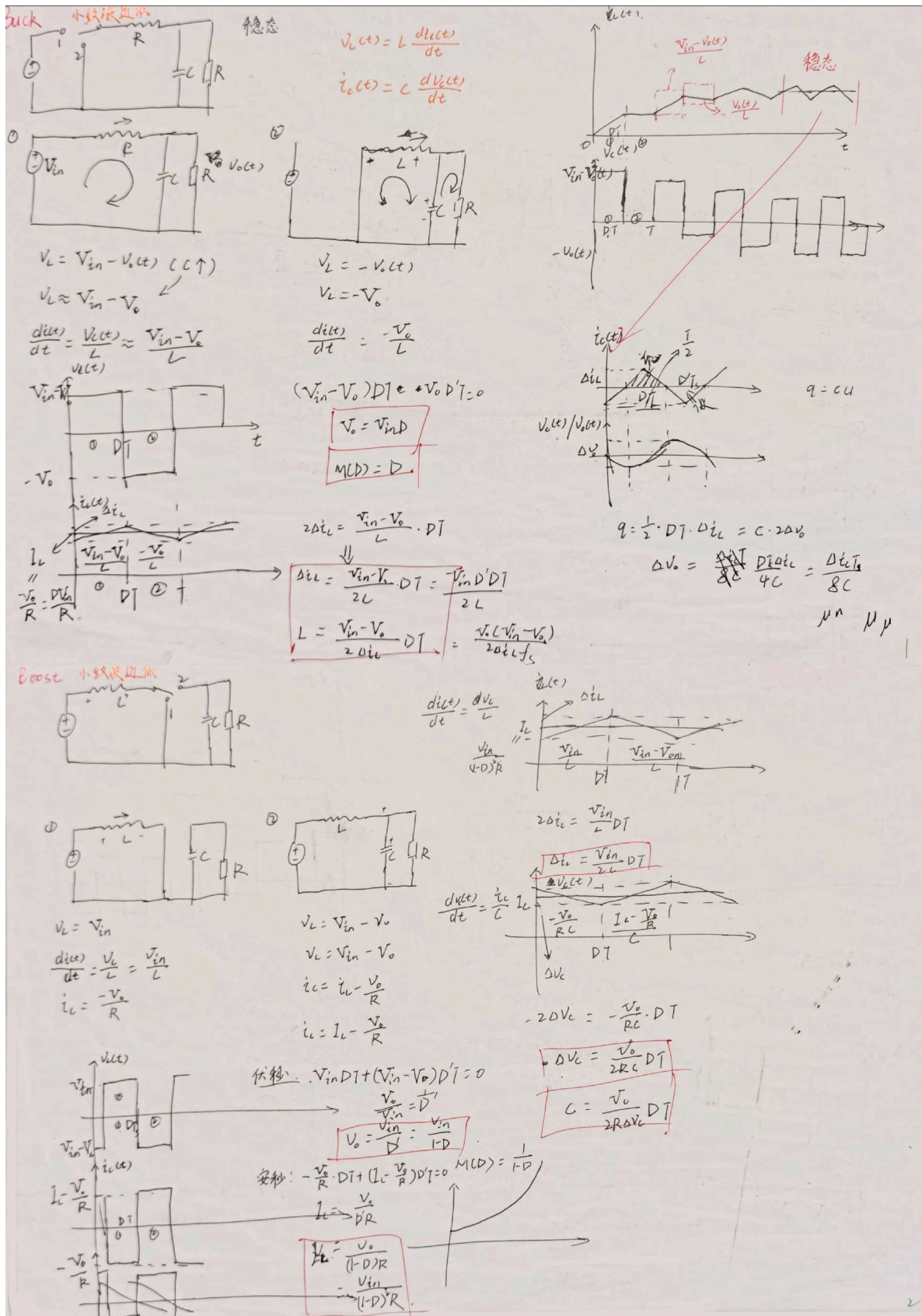
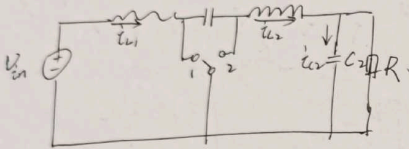


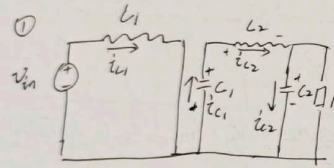
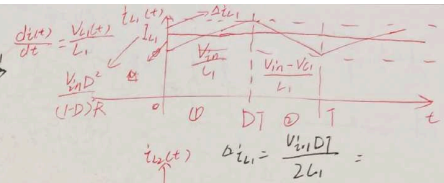
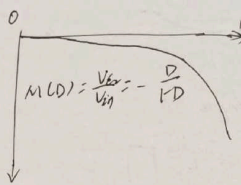
电力电子基础



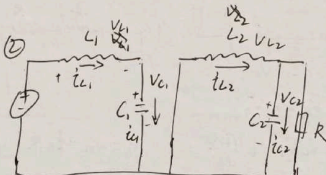
Cuk 小波逆变器



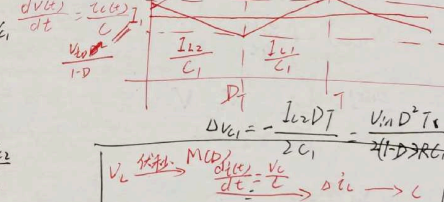
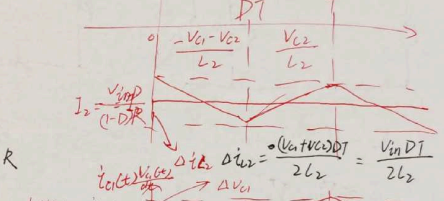
$$D = \frac{t_{on}}{T} \quad D' = \frac{t_{off}}{T} = 1-D$$



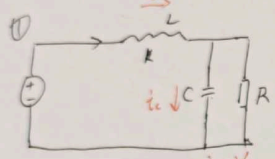
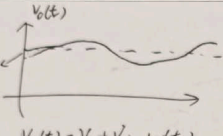
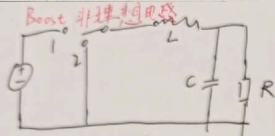
$$\begin{aligned} V_{L1} &= V_{in} & V_{C1} &= V_{in} \\ V_{L2} &= -V_{C1} - V_{C2} & V_{C2} &= -V_{C1} - V_{C2} \\ i_{L1} &= i_{L2} & i_{C1} &= i_{L2} \\ i_{C2} &= i_{L2} - \frac{V_{C2}}{R} & i_{C2} &= i_{L2} - \frac{V_{C2}}{R} \end{aligned}$$



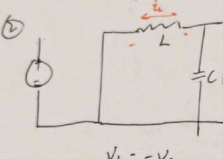
$$\begin{aligned} V_{L1} &= V_{in} - V_{C1} & V_{C1} &= V_{in} - V_{C1} \\ V_{L2} &= V_{C2} & V_{C2} &= V_{C2} \\ i_{L1} &= i_{L1} & i_{C1} &= i_{L1} \\ i_{C2} &= i_{L2} - \frac{V_{C2}}{R} & i_{C2} &= i_{L2} - \frac{V_{C2}}{R} \end{aligned}$$



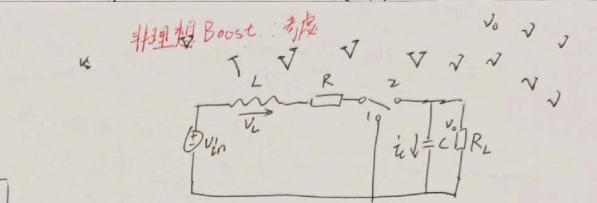
$$\begin{aligned} V_o &= \frac{V_{in}}{1-D} \\ V_{C2} &= -\frac{V_{in} D}{1-D} \\ I_2 &= \frac{V_{in} D}{(1-D) R} \\ I_1 &= \frac{V_{in} D^2}{(1-D)^2 R} \end{aligned}$$



$$\begin{aligned} V_L &= V_{in} - V_o(t) \\ V_L &\approx V_{in} - V_o \\ V_L(t) &= L \frac{di_L(t)}{dt} \end{aligned}$$



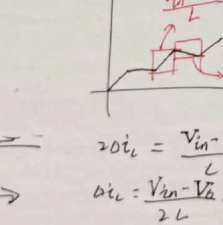
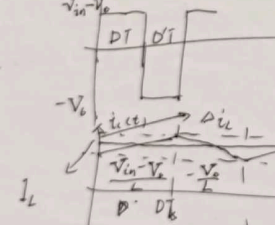
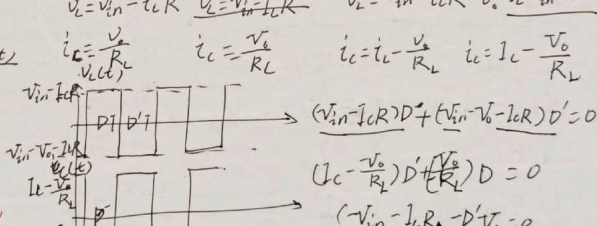
$$\begin{aligned} V_L &= -V_o \\ V_L &\approx -V_o \\ \frac{di_L(t)}{dt} &= \frac{V_L(t)}{L} \end{aligned}$$



$$\begin{aligned} V_L &= V_{in} - i_L R \\ V_L &= V_{in} - i_L R \\ i_L &= \frac{V_{in}}{R} \\ i_D &= \frac{V_o}{R_L} \end{aligned}$$

$$\frac{di_L(t)}{dt} = \frac{V_L(t)}{L} \approx \frac{V_{in} - V_o}{L}$$

$$\frac{di_L(t)}{dt} \approx -\frac{V_o}{L}$$

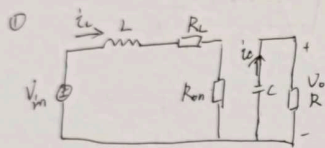
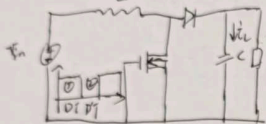


$$\begin{aligned} (V_{in} - i_L R) D + (V_{in} - V_o - i_L R) D' &= 0 \\ (i_L - \frac{V_o}{R_L}) D + (V_{in} - V_o - i_L R) D' &= 0 \\ (V_{in} - i_L R - D' V_o) &= 0 \\ D L - \frac{V_o}{R_L} &= 0 \\ \frac{V_o}{V_{in}} &= \frac{1}{D' (1 + R_L D'^2 R)} \end{aligned}$$

$$\begin{aligned} 20 i_L &= \frac{V_{in} - V_o}{L} D T \\ \Delta i_L &= \frac{V_{in} - V_o}{2L} D T \quad L = \frac{V_{in} - V_o}{20 i_L} D T \end{aligned}$$

$$\frac{V_o}{V_{in}} = \frac{1}{D' (1 + R_L D'^2 R)}$$

Buck 电路 降压 斩波电路

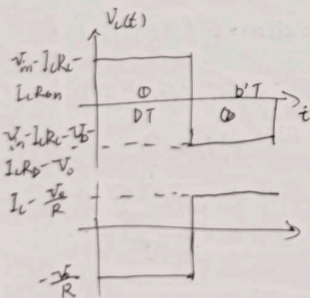


$$V_L = V_{in} - i_L R_L - i_L R_{on}$$

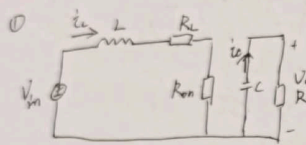
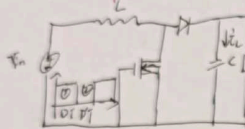
$$V_L = V_{in} - i_L R_L - i_L R_{on}$$

$$i_L = -\frac{V_o}{R}$$

$$= -\frac{V_o}{R}$$



Buck 电路 降压 斩波电路

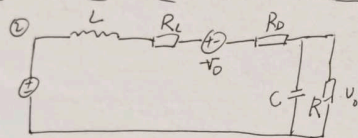
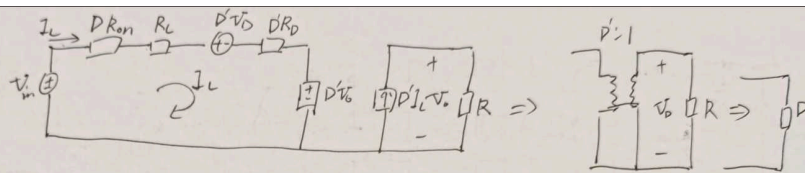
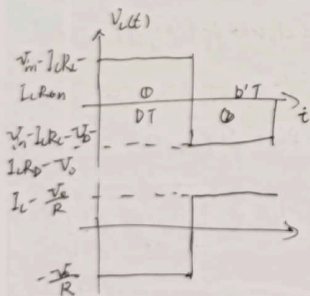


$$V_L = V_{in} - i_L R_L - i_L R_{on}$$

$$V_L = V_{in} - i_L R_L - i_L R_{on}$$

$$i_L = -\frac{V_o}{R}$$

$$= -\frac{V_o}{R}$$



$$V_L = V_{in} - i_L R_L - V_o - i_L R_D - V_o$$

$$\approx V_{in} - i_L R_L - V_o - i_L R_D - V_o$$

$$i_L = i_L - \frac{V_o}{R}$$

$$= i_L - \frac{V_o}{R}$$

$$P_{in} = (V_{in})(I_L) \quad P_{out} = (V_o)(D I_L)$$

$$\eta = D \frac{V_o}{V_{in}} = \left(1 - \frac{D' V_o}{V_{in}}\right)$$

条件 conditions for high efficiency $\left(1 + \frac{R_L + D R_{on} + D' R_D}{D^2 R}\right)$

$$\frac{V_{in}}{D'} \gg V_o$$

and

$$D'^2 R \gg R_L + D R_{on} + D' R_D$$

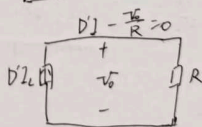
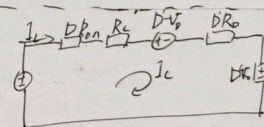
$$\text{分压思路: } \eta = \frac{1}{V_{in}} (V_{in} - D' V_o) \frac{D^2 R}{D^2 R + R_L + D R_{on} + D' R_D}$$

$$D(V_{in} - i_L R_L - i_L R_{on}) + (V_{in} - i_L R_L - V_o - i_L R_D - V_o) D' = 0$$

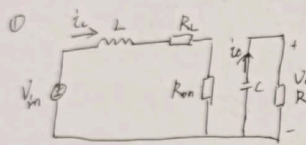
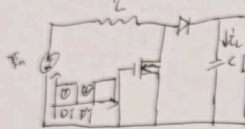
$$V_{in} - i_L R_L - i_L D R_{on} - D' V_o - D' i_L R_D - V_o D' = 0$$

$$-\frac{V_o}{R} \cdot D + \left(1 - \frac{V_o}{R}\right) D' = 0$$

$$-\frac{V_o}{R} + i_L D' = 0$$



Buck 电路 降压 斩波电路

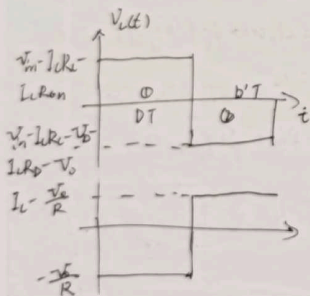


$$V_L = V_{in} - i_L R_L - i_L R_{on}$$

$$V_L = V_{in} - i_L R_L - i_L R_{on}$$

$$i_L = -\frac{V_o}{R}$$

$$= -\frac{V_o}{R}$$

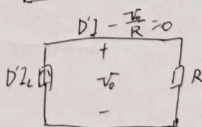
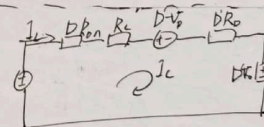


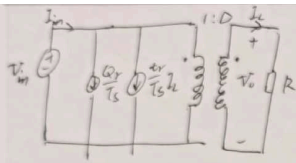
$$D(V_{in} - i_L R_L - i_L R_{on}) + (V_{in} - i_L R_L - V_o - i_L R_D - V_o) D' = 0$$

$$V_{in} - i_L R_L - i_L D R_{on} - D' V_o - D' i_L R_D - V_o D' = 0$$

$$-\frac{V_o}{R} \cdot D + \left(1 - \frac{V_o}{R}\right) D' = 0$$

$$-\frac{V_o}{R} + i_L D' = 0$$





$$P_{sw} = V_{in} \left(\frac{I_o}{f_s} + \frac{Q_r}{f_s} \right)$$

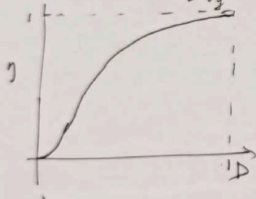
$$P_{out} = V_o I_o$$

$$P_{in} = V_{in} C D I_o + \frac{r_s}{f_s} I_o + \frac{Q_r}{f_s}$$

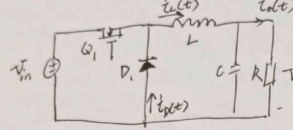
$$\eta = \frac{V_o}{V_{in}} = D$$

$$\eta = \frac{P_{out}}{P_{in}} = \frac{1}{1 + f_s \left(\frac{r_s}{D} + \frac{Q_r R}{D^2 V_o} \right)}$$

($V_{in} = 24V$, $f_s = 100kHz$, $R = 15\Omega$, $Q_r = 0.75\mu C$, $t_r = 75ns$)



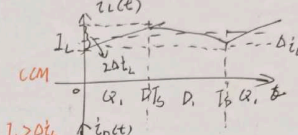
DCM (B. 续流模式) Buck



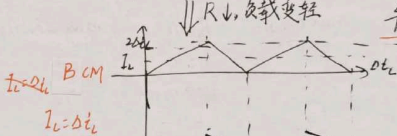
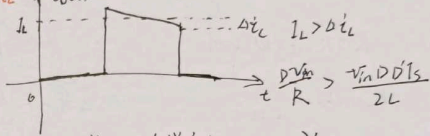
二极管最小电流: $I_L - \Delta i_L$

$$\text{电流分量 } I_L = I_o = \frac{V_o}{R} = \frac{D V_{in}}{R}$$

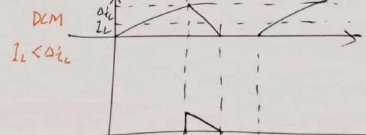
DCM 起、停与模式边界



$I_L > \Delta i_L$



$I_L = \Delta i_L$



$I_L < \Delta i_L$

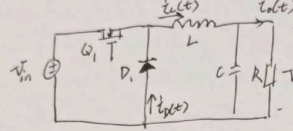
$$R < R_{crit}(D), \text{CCM} \quad K < K_{crit}(D) \text{ for DCM}$$

$$R > R_{crit}(D), \text{DCM}$$

$$K_{crit}(D) = \frac{2L}{RT_s}$$

$$K = \frac{2L}{RT_s}, K_{crit}(D) = D'$$

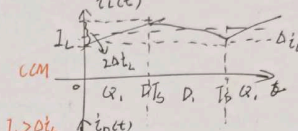
DCM (B. 续流模式) Buck



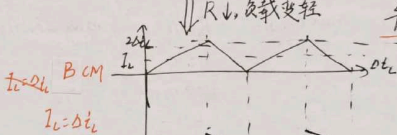
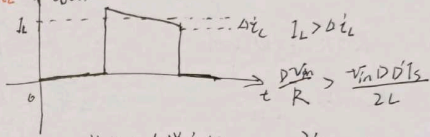
二极管最小电流: $I_L - \Delta i_L$

$$\text{电流分量 } I_L = I_o = \frac{V_o}{R} = \frac{D V_{in}}{R}$$

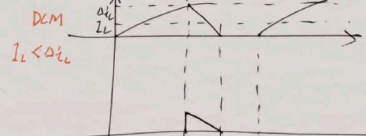
DCM 起、停与模式边界



$I_L > \Delta i_L$



$I_L = \Delta i_L$



$I_L < \Delta i_L$

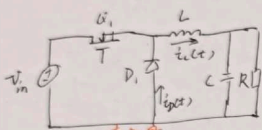
$$R < R_{crit}(D), \text{CCM} \quad K < K_{crit}(D) \text{ for DCM}$$

$$R > R_{crit}(D), \text{DCM}$$

$$K_{crit}(D) = \frac{2L}{RT_s}$$

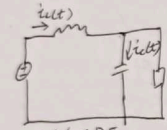
$$K = \frac{2L}{RT_s}, K_{crit}(D) = D'$$

Buck 电压增益分析. DCM 求解



由于工作在 DCM, 不能使用小纹波近似

subinterval 1

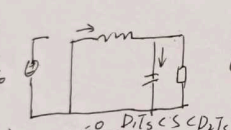


$$0 < t < D_1 T_s$$

$$v_L(t) = V_{in} - V_o$$

$$i_L(t) = i_L(t) - \frac{V_o(t)}{R}$$

subinterval 2

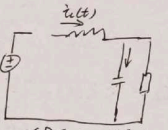


$$D_1 T_s < t < D_1 T_s + D_2 T_s$$

$$v_L(t) = -V_o(t)$$

$$i_L(t) = i_L(t) - \frac{V_o(t)}{R}$$

subinterval 3



$$D_1 T_s + D_2 T_s < t < T_s$$

$$v_L(t) = 0, i_L = 0$$

$$i_L(t) = i_L(t) - \frac{V_o(t)}{R}$$

Small ripple approximation

$$V_L(t) \approx V_{in} - V_o$$

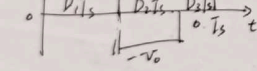
$$i_L(t) \approx i_L(t) - \frac{V_o}{R}$$

$$V_L(t) \approx -V_o(t)$$

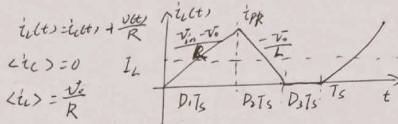
$$i_L(t) \approx i_L(t) - \frac{V_o(t)}{R}$$

$$V_L(t) = 0$$

$$i_L(t) = -\frac{V_o}{R}$$



$$V_o = V_{in} \frac{D_1}{D_1 + D_2}$$

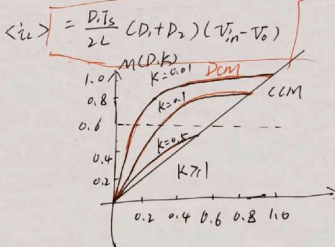


$$i_{L,CD}(T_s) = i_{pk} = \frac{V_{in} - V_o}{L} \cdot D_1 T_s$$

$$\langle i_L \rangle = \frac{1}{T_s} \int_0^{T_s} i_L(t) dt$$

$$= \frac{1}{T_s} \cdot \frac{1}{2} \cdot i_{pk} (D_1 + D_2) T_s$$

$$= \frac{1}{T_s} \cdot \frac{1}{2} \cdot \frac{V_{in} - V_o}{L} \cdot D_1 T_s \cdot (D_1 + D_2) T_s$$



$$K = \frac{2L}{RT_s} \cdot K_{crit}(D) = D'$$

$$M = \begin{cases} D & K > K_{crit} \\ \frac{2}{1 + \sqrt{1 + \frac{4K}{D^2}}} & K < K_{crit} \end{cases}$$

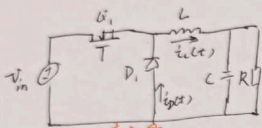
Eliminate D_2 , solve for V_o

$$\frac{V_o}{V_{in}} = \frac{2}{1 + \sqrt{1 + \frac{4K}{D^2}}}$$

$$K = \frac{2L}{RT_s}$$

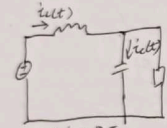
$K < K_{crit}$ (DCM)

Buck 电压增益分析. DCM 求解



由于工作在 DCM, 不能使用小纹波近似

subinterval 1

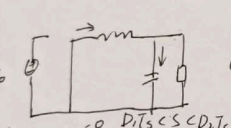


$$0 < t < D_1 T_s$$

$$v_L(t) = V_{in} - V_o$$

$$i_L(t) = i_L(t) - \frac{V_o(t)}{R}$$

subinterval 2

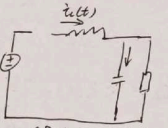


$$D_1 T_s < t < D_1 T_s + D_2 T_s$$

$$v_L(t) = -V_o(t)$$

$$i_L(t) = i_L(t) - \frac{V_o(t)}{R}$$

subinterval 3



$$D_1 T_s + D_2 T_s < t < T_s$$

$$v_L(t) = 0, i_L = 0$$

$$i_L(t) = i_L(t) - \frac{V_o(t)}{R}$$

Small ripple approximation

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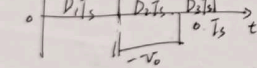
$$i_L(t) \approx i_L(t) - \frac{V_o}{R}$$

$$V_L(t) \approx -V_o(t)$$

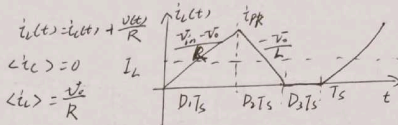
$$i_L(t) \approx i_L(t) - \frac{V_o(t)}{R}$$

$$V_L(t) = 0$$

$$i_L(t) = -\frac{V_o}{R}$$



$$V_o = V_{in} \frac{D_1}{D_1 + D_2}$$

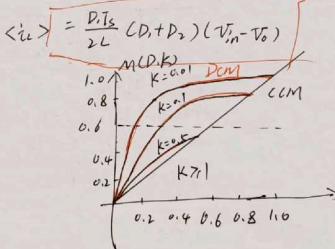


$$i_{L,CD}(T_s) = i_{pk} = \frac{V_{in} - V_o}{L} \cdot D_1 T_s$$

$$\langle i_L \rangle = \frac{1}{T_s} \int_0^{T_s} i_L(t) dt$$

$$= \frac{1}{T_s} \cdot \frac{1}{2} \cdot i_{pk} (D_1 + D_2) T_s$$

$$= \frac{1}{T_s} \cdot \frac{1}{2} \cdot \frac{V_{in} - V_o}{L} \cdot D_1 T_s \cdot (D_1 + D_2) T_s$$



$$K = \frac{2L}{RT_s} \cdot K_{crit}(D) = D'$$

$$M = \begin{cases} D & K > K_{crit} \\ \frac{2}{1 + \sqrt{1 + \frac{4K}{D^2}}} & K < K_{crit} \end{cases}$$

Eliminate D_2 , solve for V_o

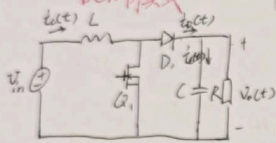
$$\frac{V_o}{V_{in}} = \frac{2}{1 + \sqrt{1 + \frac{4K}{D^2}}}$$

$$K = \frac{2L}{RT_s}$$

$K < K_{crit}$ (DCM)

Boost 电路下压电压增益分析

DCM 模式



$$I_L > I_{Lc} \text{ CCM}$$

$$I_L < I_{Lc} \text{ DCM}$$

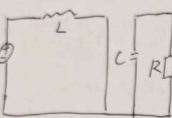
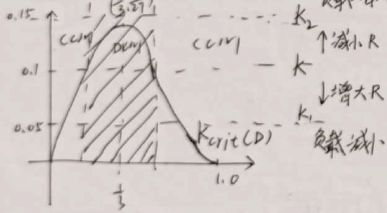
$$\frac{V_{in}}{(1-D)^2 R} > \frac{D^2 V_{in}}{2L} \text{ CCM}$$

$$K_s \frac{2L}{RT_s} > DD'^2 \text{ CCM}$$

$$K > K_{crit}(D) \text{ CCM}$$

$$K < K_{crit}(D) \text{ DCM}$$

$$K = \frac{2L}{RT_s}, K_{crit}(D) = DD'^2$$

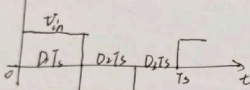


$$V_L(t) = V_{in}$$

$$i_L(t) = -\frac{V_o(t)}{R}$$

$$V_L(t) \approx V_{in}$$

$$i_L(t) \approx -\frac{V_o}{R}$$



$$D_1 V_{in} + D_2 (V_{in} - V_o) + D_3 (0) = 0$$

$$V_o = \frac{D_1 + D_2}{D_2} V_{in}$$

$$V_o = \frac{D_1 + D_2}{D_2} V_{in}$$

$$V_o = \frac{D_1 + D_2}{D_2} V_{in}$$

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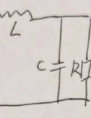
$$V_o = \frac{D_1 + D_2}{D_2} V_{in}$$

$$V_o = \frac{D_1 + D_2}{D_2} V_{in}$$

$$V_o = \frac{D_1 + D_2}{D_2} V_{in}$$

$$V_o = \frac{D_1 + D_2}{D_2} V_{in}$$

DCM



$$V_L(t) = V_{in} - V_o(t)$$

$$i_L(t) = i_L(t) - \frac{V_o(t)}{R}$$

$$V_L(t) \approx V_{in} - V_o$$

$$i_L(t) \approx i_L(t) - \frac{V_o}{R}$$

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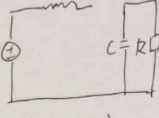
$$i_L(t) \approx i_L(t) - \frac{V_o}{R}$$

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$$i_L(t) \approx i_L(t) - \frac{V_o}{R}$$



$$V_L(t) = 0, i_L = 0$$

$$i_L(t) = -\frac{V_o(t)}{R}$$

$$V_L(t) \approx 0, i_L = 0$$

$$i_L(t) \approx -\frac{V_o}{R}$$

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$$i_L(t) \approx -\frac{V_o}{R}$$

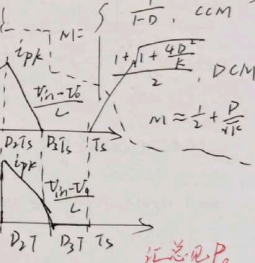
$$i_L(t) \approx -\frac{V_o}{R}$$

$$D_2 = D_1 \frac{V_{in}}{V_o - V_{in}}$$

$$V_o^2 - V_o V_{in} - \frac{V_{in}^2 D_1^2}{K} = 0$$

$$\frac{V_o}{V_{in}} = M(D, K) = \frac{1 + \sqrt{1 + 4 \frac{D_1^2}{K}}}{2}$$

$$K = \frac{2L}{RT_s} \cdot K_{crit}(D) = DD'^2$$



$$i_{pk} = \frac{V_{in}}{L} D_1 T_s$$

$$\langle i_D \rangle = \frac{1}{T_s} \int_0^{T_s} i_D(t) dt$$

$$= \frac{1}{T_s} \cdot \frac{1}{2} \cdot i_{pk} D_2 T_s$$

$$= \frac{1}{T_s} \cdot \frac{1}{2} \cdot \frac{V_{in}}{L} D_1 T_s \cdot D_2 T_s$$

$$= \frac{V_{in} D D_2 T_s}{2L} = \frac{V_o}{R}$$

电感模型

法拉第定律

$$V_{ind}(t) = \frac{d\Phi(t)}{dt}$$

$$V(t) = n \frac{d\Phi(t)}{dt}$$

$$= n \frac{d}{dt} \left(\frac{\Phi(t)}{dt} \right)$$

$$= n A_c \frac{dB(t)}{dt}$$

$$= n A_c \frac{dB(t)}{dt}$$

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$$= n A_c \frac{dB(t)}{dt}$$

$$= n A_c \frac{dB(t)}{dt}$$

Magnetic circuits

磁路

$$F = Hl$$

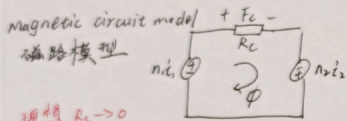
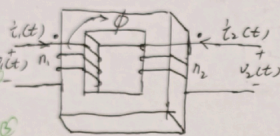
$$= \frac{l}{\mu A_c} \Phi$$

变压器模型

$$R_c = \frac{l_m}{\mu_0 \mu_r A_c}$$

$$F_c = n_1 i_1 + n_2 i_2$$

$$\phi R_c = n_1 i_1 + n_2 i_2$$



理想 $R_c \rightarrow 0$

$$MMF F_c = \phi R_c$$

$$0 = n_1 i_1 + n_2 i_2$$

$$v_1 = n_1 \frac{d\phi}{dt} \quad v_2 = n_2 \frac{d\phi}{dt}$$

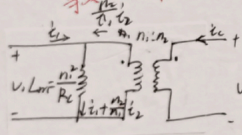
$$\frac{d\phi}{dt} = \frac{v_1}{n_1} = \frac{v_2}{n_2}$$

并理想

$$\phi R_c = n_1 i_1 + n_2 i_2$$

$$v_1 = \frac{n_1}{R_c} \frac{d}{dt} \left[\frac{1}{n_1} + \frac{n_2}{n_1} i_2 \right]$$

等效电路模型



$$v_2(t) = L \frac{di_2(t)}{dt}$$

$$i_{mp}(t) = \frac{1}{L_{mp}} \int v_1(t) dt$$

$$v_1(t) = n_1 A_c \frac{d\phi(t)}{dt}$$

$$\phi(t) = \frac{1}{n_1 A_c} \int v_1(t) dt$$

越大越易饱和

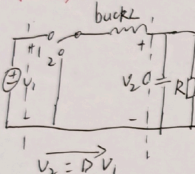
$$\lambda_1 = \int_{t_1}^{t_2} v_1(t) dt$$

增大磁芯面积与横面积,可以降低磁通密度,防止饱和。

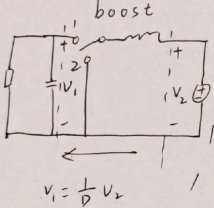
12

功率变换器行变

反向往复:

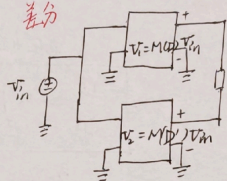


$$V_2 = D V_1$$



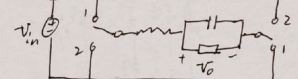
$$V_1 = \frac{1}{D} V_2$$

差分

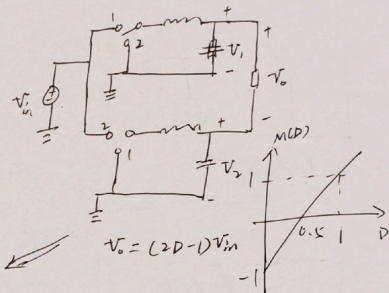
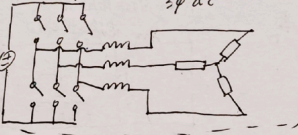


$$V_0 = V_1 - V_2$$

直流交流全桥逆变器



直流三相交流逆变器



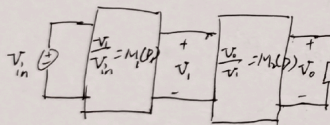
$$V_0 = (2D - 1) V_{in}$$

$$\frac{-V_0}{V_{in}} = -\frac{D}{1-D} V_{in}$$

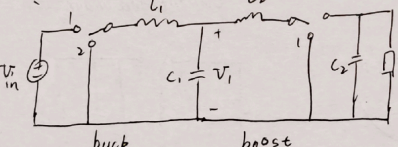
输出极性与下面相反

buck-boost

级联

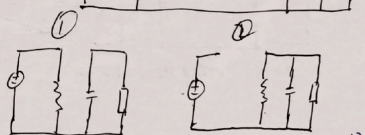


$$\frac{V_0}{V_{in}} = M_1(D) M_2(D)$$



$$\frac{V_0}{V_{in}} = D \quad \frac{V_0}{V_1} = \frac{1}{1-D}$$

$$\frac{V_0}{V_{in}} = \frac{D}{1-D}$$

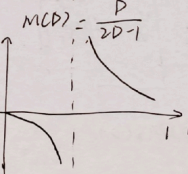
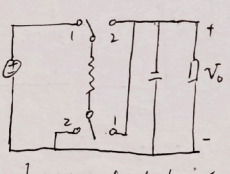
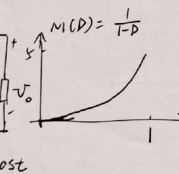
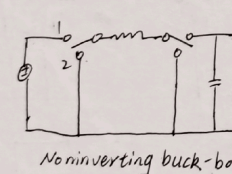
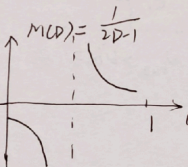
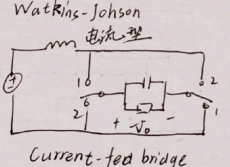
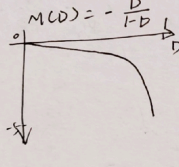
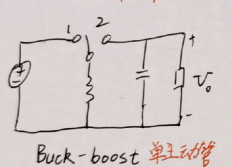
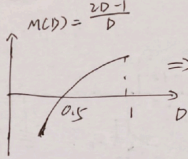
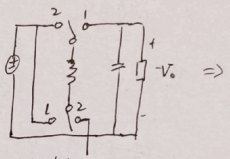
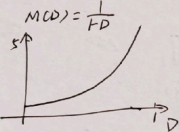
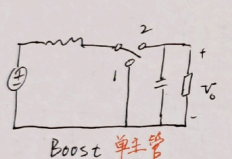
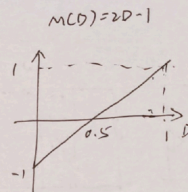
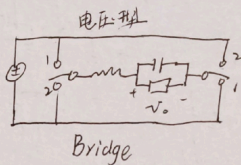
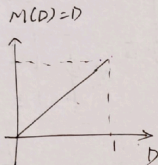
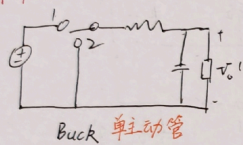


13

DC-DC 变换器

非隔离

单电感

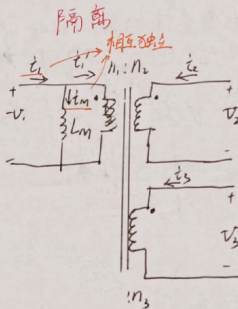
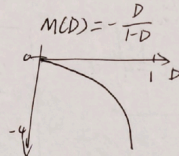
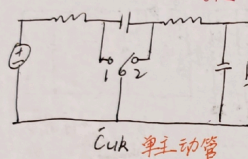


14

DC-DC 变换器

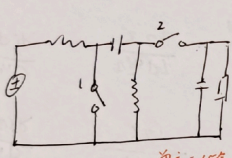
非隔离

双电感



$$\frac{V_1}{n_1} = \frac{V_2}{n_2} = \frac{V_3}{n_3}$$

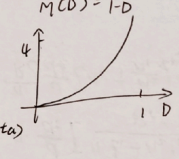
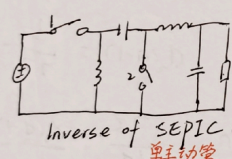
$$0 = n_1 i_1 + n_2 i_2 + n_3 i_3$$



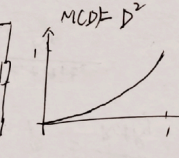
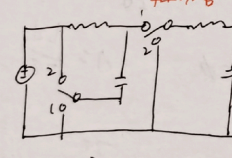
$$V_L(t) = L_M \frac{di_m(t)}{dt}$$

$$i_m(t) - i_m(0) = \frac{1}{L_M} \int_0^t V_L(t) dt$$

所以励磁电流是由施加在绕组上的电压积分决定的。故 V 的直流必须为零



$$0 = \frac{1}{T_s} \int_0^{T_s} V_L(t) dt$$

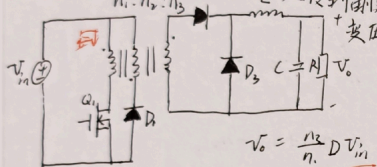


Buck

15

DC-DC 变换器 Forward Converter

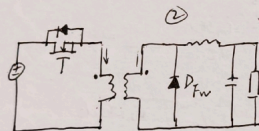
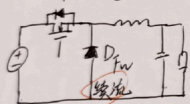
Buck 隔离 正激变换器 (开关管导通时, 能量直接由原边传到副边, 变压器只起变压和隔离作用)



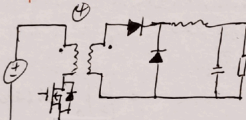
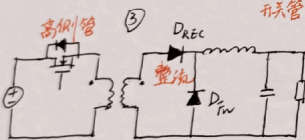
$$V_o = \frac{n_2}{n_1} D V_{in}$$

Buck 正激

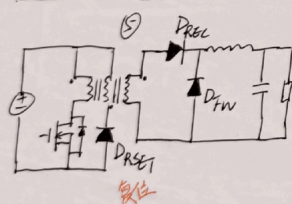
推演 ①



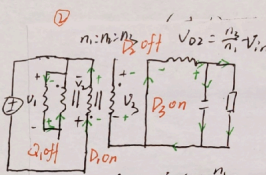
开关管导通, 输出电容会倒灌电流



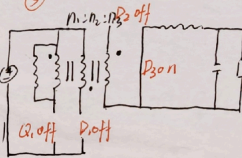
低侧管电压器饱和



复位



$$V_{Q1} = V_{in} (1 + \frac{n_1}{n_2})$$



$$\langle V_{Q1} \rangle = D C V_{in} + D_2 C (-\frac{n_1}{n_2} V_{in}) + D_3 (0) = 0$$

$$D_2 = \frac{n_2}{n_1} D$$

$$D_3 = 1 - D - D_2 \geq 0$$

$$D_3 = 1 - D (1 + \frac{n_1}{n_2}) \geq 0$$

$$D \leq \frac{1}{1 + \frac{n_1}{n_2}} \cdot n_1 = n_2 \cdot D \leq \frac{1}{2}$$

空载比过大, 在一个周期内励磁电感净流量不为零

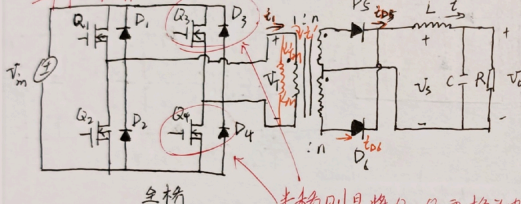
16

DC-DC 变换器 正激

Full Bridge Isolated Buck Converters

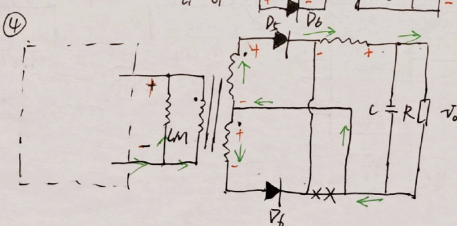
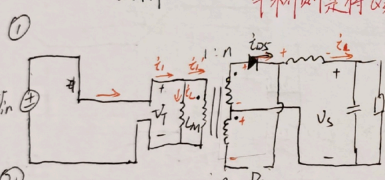
全桥式隔离降压变换器

全桥 Full



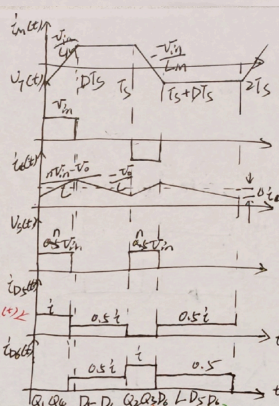
全桥

半桥则是将 Q2, Q4 更换为电容



$$V_L + V_o = V_s$$

$$V_L = -V_o$$



- ① Q2, Q3 两端均为 V_{in}
D2 两端电压为 $2nV_{in}$
- ② Q1, Q4 两端电压为 $2V_{in}$
Lm 感应出上负下正的 V_{in}
- ③ Q1, Q4 两端均为 V_{in}
D5 两端电压为 $2nV_{in}$
- ④ Q2, Q3 两端为 $2V_{in}$
Lm 感应出上正下负的 V_{in}

$$(nV_{in} - V_o)D + (-\frac{V_o}{n})D = 0$$

$$V_o < V_s$$

$$V_o = n D V_{in}$$

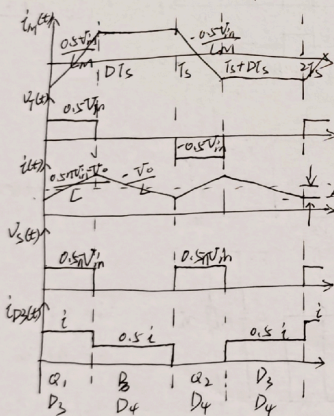
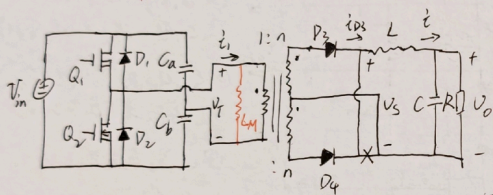
$$m(D) = nD$$

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DC-DC 变换器

Half-Bridge Isolate Buck Converter

半桥隔离降压变换器

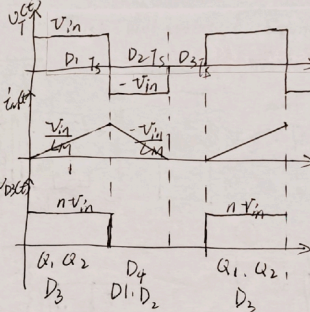
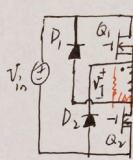


伏秒: $(\frac{1}{2} V_{in} - V_o) D + (C - V_o) D' = 0$

$V_o = n \cdot D \cdot V_{in}$

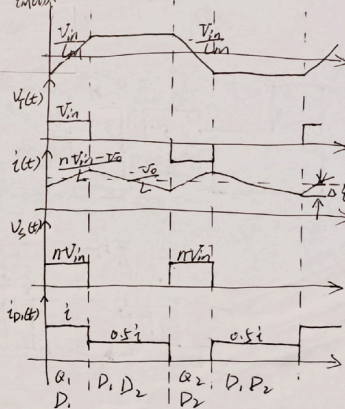
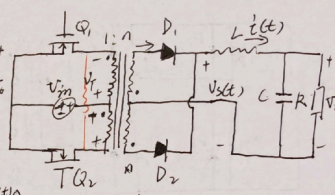
TWO-transistor forward converter

双管正激变换器



Push-pull isolate buck converter

推挽隔离降压变换器

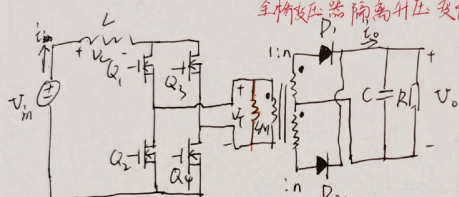


18

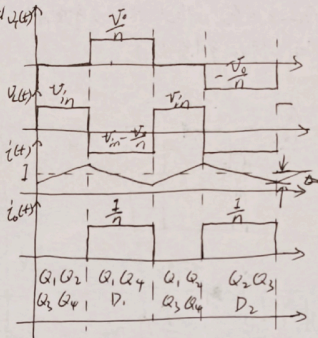
DC-DC 变换器

隔离 Boost 正激

Full-bridge transformer-isolated boost converter
全桥变压器隔离升压变换器

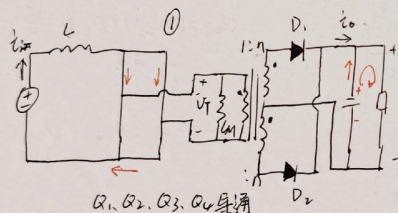


电流源型

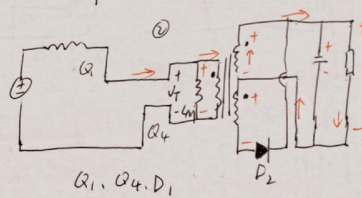


$\langle V_L \rangle = D C V_{in} + D' C V_{in} - \frac{V_o}{n} = 0$

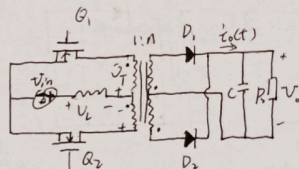
$M(D) = \frac{V_o}{V_{in}} = \frac{n}{D}$



Q1, Q2, Q3, Q4 导通

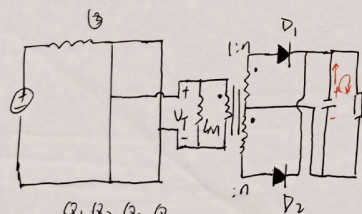


Q1, Q4, D1

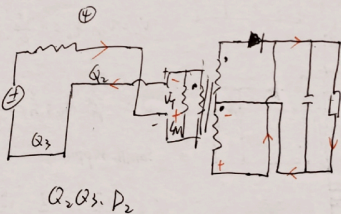


$\langle V_L \rangle = D C V_{in} + D' C V_{in} - \frac{V_o}{n} = 0$

$M(D) = \frac{V_o}{V_{in}} = \frac{n}{D}$



Q1, Q2, Q3, Q4



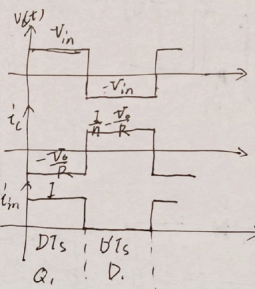
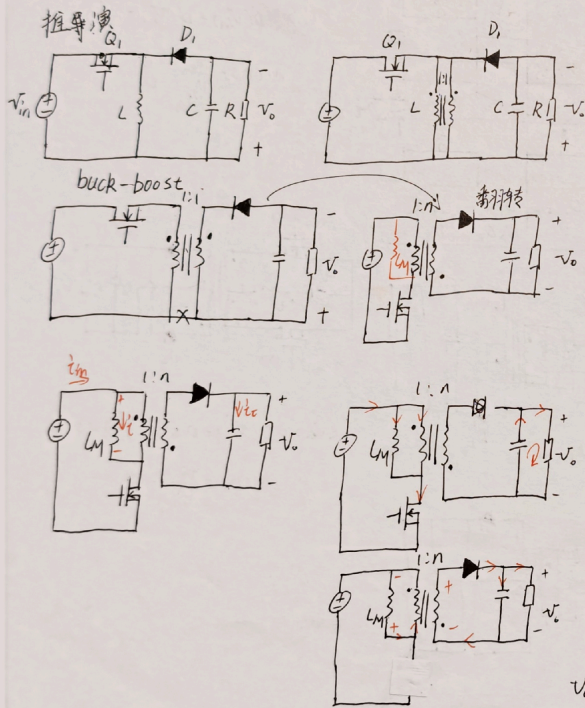
Q2, Q3, D2

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DC-DC 转换器

反激式转换器 (开关管导通时, 能量先存在变压器磁芯里, 开关管关断时, 能量从磁芯释放到副边)

Flyback Converter 关断时, 磁芯储能释放到副边)



伏秒: $\langle V_c \rangle = DC \cdot V_{in} + D'(-\frac{V_o}{n}) = 0$

$M(D) = \frac{V_o}{V_{in}} = n \frac{D}{D'}$

荷电: $\langle i_c \rangle = D(\frac{V_o}{R}) + D'(\frac{1}{n} - \frac{V_o}{R}) = 0$

$I = \frac{nV_o}{D'R}$

$I_{in} = \langle i_{in} \rangle = D(I) + D'(0)$

CCM:

$V_L = V_{in}, i_c = -\frac{V_o}{R}, i_{in} = I$

小纹波近似

$V_L = V_{in}, i_c = -\frac{V_o}{R}, i_{in} = I$

$V_L = -\frac{V_o}{n}, i_c = \frac{I}{n} - \frac{V_o}{R}, i_{in} = 0$

Small ripple

$V_L = -\frac{V_o}{n}, i_c = \frac{I}{n} - \frac{V_o}{R}, i_{in} = 0$

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交流电的基本概念

Basic AC concept

Fourier series - periodic functions

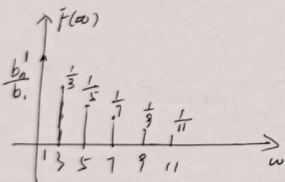
$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(n\omega t) + b_n \sin(n\omega t)] \quad (\omega = \frac{2\pi}{T})$

$= \frac{a_0}{2} + a_1 \cos(\omega t) + a_2 \cos(2\omega t) + \dots$
 $+ b_1 \sin(\omega t) + b_2 \sin(2\omega t) + \dots$

$a_n = \frac{2}{T} \int_0^T \cos(n\omega t) f(t) dt$

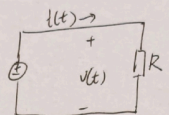
$b_n = \frac{2}{T} \int_0^T \sin(n\omega t) f(t) dt$

$f(t) = \begin{cases} 1, & 0 < t < \frac{T}{2} \\ -1, & \frac{T}{2} < t < T \end{cases}$



谐波总增加有效值, 但不一定能加平均功率。

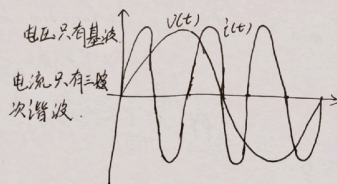
损耗 ↑



$v(t) = V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega t - \varphi_n)$
 $i(t) = I_0 + \sum_{n=1}^{\infty} I_n \cos(n\omega t - \theta_n)$

$W = \int_0^T v(t) i(t) dt$

平均功率: $P_{av} = \frac{W}{T} = \frac{1}{T} \int_0^T v(t) i(t) dt$



瞬时功率

平均功率为零, 但瞬时功率不为零。

时域有效值: (rms value) $= \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$

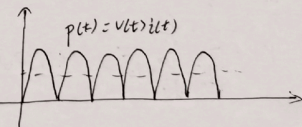
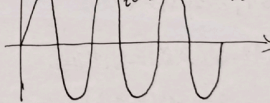
频域有效值: (rms value) $= \sqrt{V_0^2 + \sum_{n=1}^{\infty} \frac{V_n^2}{2}}$

"相同频率的电压电流才有功率" (不严谨)

$= \frac{1}{T} \int_0^T [V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega t - \varphi_n)] [I_0 + \sum_{n=1}^{\infty} I_n \cos(n\omega t - \theta_n)] dt$
 $= \begin{cases} 0, & n \neq m \\ \frac{V_n I_n}{2} \cos(\varphi_n - \theta_n), & n = m \end{cases}$

$= V_0 I_0 + \sum_{n=1}^{\infty} \frac{V_n I_n}{2} \cos(\varphi_n - \theta_n)$

均为三次谐波。



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power factor = $\frac{\text{average power}}{(\text{rms voltage})(\text{rms current})}$
功率因数 = $\frac{\text{有功功率}}{\text{有效电压} \times \text{有效电流}}$

视在功率: 有效电压与有效电流乘积, $S = |S| = \frac{1}{2} V_{rms} I_{rms}$

平均(有功)功率: $P = \operatorname{Re}[\frac{1}{2} V I^*]$

无功功率: $Q = \operatorname{Im}[\frac{1}{2} V I^*]$

线性电阻负载, 非正弦电压

$I_n = \frac{V_n}{R}$

(rms current) = $\sqrt{I_0^2 + \sum_{n=1}^{\infty} \frac{I_n^2}{2}}$
 $= \sqrt{\frac{V_0^2}{R^2} + \sum_{n=1}^{\infty} \frac{V_n^2}{2R^2}}$

$= \frac{1}{R} (\text{rms voltage}) P_{av}$

$= V_0 I_0 + \sum_{n=1}^{\infty} \frac{V_n I_n}{2} \cos(\varphi_n - \theta_n)$

此时功率因数为1.

非线性动态负载, 正弦电压

$P_{av} = \frac{V_1 I_1}{2} \cos(\varphi_1 - \theta_1)$

$PF = \frac{\frac{I_1^2}{2}}{\sqrt{I_0^2 + \sum_{n=1}^{\infty} \frac{I_n^2}{2}}} \cos(\varphi_1 - \theta_1)$

仅对正弦电压条件不成立.

功率因数 = $\frac{\frac{I_1^2}{2}}{\sqrt{I_0^2 + \sum_{n=1}^{\infty} \frac{I_n^2}{2}}} = \frac{\text{基波电流有效值}}{\text{总电流有效值}} = \frac{1}{\sqrt{1 + \text{THD}^2}}$

总谐波失真
Related to Total Harmonic Distortion (THD):

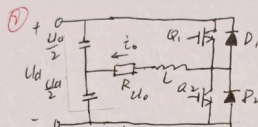
$\text{THD} = \frac{\sqrt{\sum_{n=2}^{\infty} I_n^2}}{I_1}$

= 畸变因数 $\times$ 位移因数 (distortion factor) $\cdot$ (displacement factor)

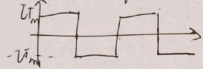
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逆变电路

Square Wave Inverters
方波逆变器



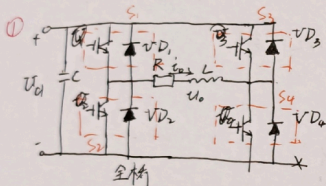
半桥



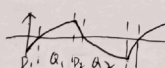
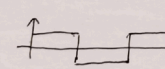
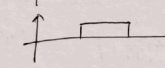
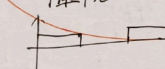
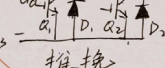
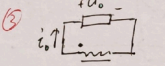
$\text{THD} = \frac{\sqrt{\sum_{n=2}^{\infty} (V_{n,rms})^2}}{V_{1,rms}} = \frac{\sqrt{V_{rms}^2 - V_{1,rms}^2}}{V_{1,rms}}$

基波占所有分量的比重

Voltage source inverters (VSI)
电压型逆变电路



全桥



输出电压(方波)傅里叶级数展开

$U_o(t) = \frac{4U_d}{\pi} (\sin \omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{5} \sin 5\omega t + \dots)$

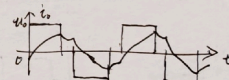
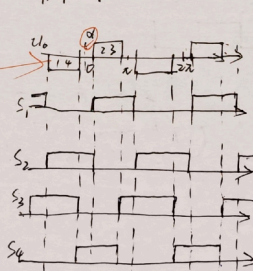
输出电压基波分量幅值

$U_1 = \frac{4U_d}{\pi} = 1.27 U_d$

基波有效值

$U_{1,rms} = \frac{U_1}{\sqrt{2}} = 0.9 U_d$

移相调压



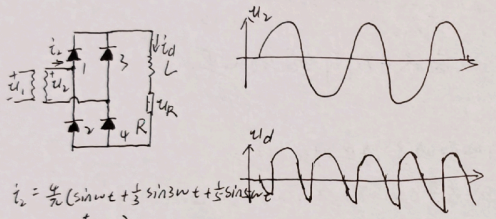
$V_o(t) = \sum_{n \text{ odd (奇)}} V_n \sin(n\omega t)$

$V_n = \frac{4V_{dc}}{n\pi} \cos(n\alpha)$

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可控整流器谐波

Harmonics of controlled rectifier



$$i_d = \frac{4}{\pi} (\sin \omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{5} \sin 5\omega t + \dots)$$

$$= \frac{4}{\pi} I_d \sum_{n=1,3,5,\dots} \frac{1}{n} \sin n\omega t$$

$$= \sum_{n=1,3,5,\dots} I_n \sin n\omega t$$

$$I_n = \frac{2\sqrt{2}I_d}{n\pi}, n=1,3,5,\dots$$

畸变因数:

$$D = \frac{I_1}{I} = \frac{2\sqrt{2}}{\pi} \approx 0.9$$

位移因数

$$\lambda_1 = \cos \varphi_1 = \cos \alpha$$

功率因数

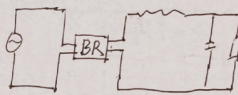
$$\lambda = \lambda_1 = \frac{2}{\pi} \cos \varphi_1$$

$$= \frac{2\sqrt{2}}{\pi} \cos \alpha$$

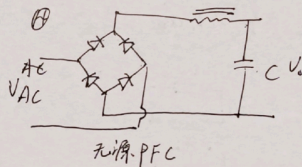
$$\approx 0.9 \cos \alpha$$

不可控整流器谐波

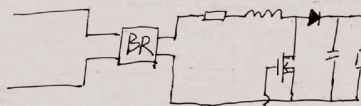
Harmonics of uncontrolled rectifier



$L \rightarrow \infty$ 时, 电流接近正弦。



无源 PFC



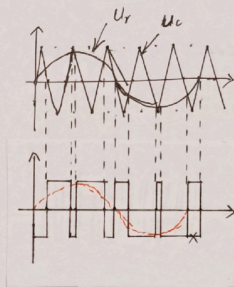
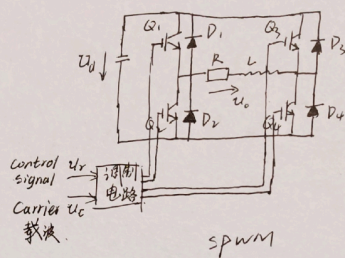
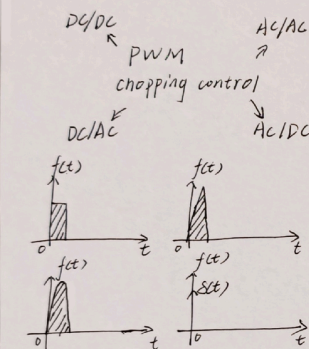
有源 PFC

controller

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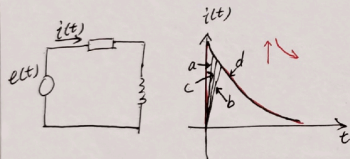
Pulse-Width Modulated Inverter

脉冲宽度调制逆变器



$$U_r > U_c \quad Q_1, Q_4$$

$$U_r < U_c \quad Q_2, Q_3$$



冲量相同的各种脉冲中的
响应波形

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逆变电路

Current source inverter (CSI)

电流型逆变电路

